Example

Problem

- ❑ Show how to multiply the complex numbers $\mathrm{a}+\mathrm{b}i$ and $\mathrm{c}+\mathrm{d}i$ using only **three multiplications** of real numbers. The algorithm should take a, b, c, and d as input and produce the real component $\mathrm{ac}-\mathrm{bd}$ and the imaginary component $\mathrm{ad}+\mathrm{bc}$ separately.

- Hints: Consider Strassen's approach
  - 1. Find three multiplications, e.g., A, B, and C
  - 2. Compute the real and imaginary components using A, B, and C

$$(\boldsymbol{a}+\boldsymbol{b}\boldsymbol{i})(\boldsymbol{c}+\boldsymbol{d}\boldsymbol{i})=(\boldsymbol{ac}-\boldsymbol{bd})+(\boldsymbol{ad}+\boldsymbol{bc})\boldsymbol{i}$$

$$A=(a+b)(c+d)=ac+ad+bc+bd$$
$$B=ac$$
$$C=bd$$

Then, the result is $(B-C)+(A-B-C)i$